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DSC 623 - Project - Logical Design

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A. Foreign Keys

a. Outlet(<u>OutletID</u> , Address, PhoneNumber, FaxNumber)
b. Vehicle(<u>RegistrationNumber</u>, OutletID (FK), Model, Make, EngineSize, Capacity, CurrentMileage, DailyHireRate, CurrentLocation)
c. Client(<u>ClientID</u>, FirstName, LastName, Address, PhoneNumber, DateOfBirth, DrivingLicenseNumber)
d. Staff(<u>StaffID</u>, OutletID (FK), FirstName, LastName, Address, PhoneNumber, DateOfBirth, Sex, DateJoined, JobTitle, Salary)
e. HireAgreement(<u>HireID</u>, RegistrationNumber (FK), ClientID (FK), StartDate, EndDate, PreMileage, PostMileage)

B. Logical Model Normalization - 3NF

- 1NF - All of the tables have non-repeating values, no multivalued attributes, each table has one primary key, and all values are atomic. Thus, our database is in 1NF.
- 2NF - As there are no composite primary keys, every non-key attribute is dependent on the whole primary key. Thus, there are no partial dependencies. So, the database is in 2NF.
- 3NF - All attributes depend on the primary key and no other attribute, so all of the tables are in 3NF.

C. Create five queries involving more than one entity (explain steps)

1. Retrieve all the staff members together with the outlet to which they are assigned, including the outlet number and staff details.
    a. Entities Involved:
        i. Staff
        ii. Outlet
    b. Steps to get result:
        i. Read Staff relation to get details of each staff member.
        ii. Read the Outlet relation to get outlet information.
        iii. Join Staff and Outlet using <u>OutletID</u>, since <u>OutletID</u> in Staff is a foreign key referencing Outlet.
        iv. Return the staff details along with the corresponding outlet information.

2. Get all the clients who have made more than one hire agreement, together with the vehicles they hired.
    a. Entities Involved:
        i. Client
        ii. HireAgreement
        iii. Vehicle
    b. Steps to get result:

i. Read the HireAgreement relation.
ii. Group the records by ClientID.
iii. Count the number of hire agreements for each client.
iv. Keep only those clients whose count is greater than one.
v. Join the filtered results with the Client relation using ClientID to get client details.
vi. Join the result with Vehicle relation using RegistrationNumber to get the details of the vehicles hired.
vii. Return the client details together with the corresponding hired vehicles.

3. Retrieve the vehicle that has been hired most frequently, based on the number of hire agreements associated with it.
   a. Entities Involved:
      i. Vehicle
      ii. HireAgreement
   b. Steps to get result:
      i. Read the HireAgreement relation.
      ii. Group the records by the RegistrationNumber.
      iii. Count the number of hire agreements associated with each vehicle.
      iv. Sort the grouped results in descending order of the hire count.
      v. Select the vehicle with the highest count.
      vi. Join the result with the Vehicle relation using the RegistrationNumber to display the full vehicle details.
      vii. Return the vehicle and its total number of hires.

4. Retrieve the hire history of a specific vehicle, including the client who hired it and the start and end dates of each hire agreement.
   a. Entities Involved:
      i. Vehicle
      ii. HireAgreement
      iii. Client
   b. Steps to get result:
      i. Read the Vehicle relation and identify the required vehicle using its RegistrationNumber.
      ii. Read the HireAgreement relation and select all the hire agreements associated with that RegistrationNumber.
      iii. Join the filtered hire agreements with the Client relation using ClientID to get client details.
      iv. Return the vehicle, the client who hired it, and the corresponding hire start and end dates.

5. Get all the vehicles currently assigned to each outlet, including the vehicle registration number, make, and model.
    a. Entities Involved:
        i. Vehicle
        ii. Outlet
    b. Steps to get result:
        i. Read the Vehicle relation to get the vehicle details.
        ii. Read the Outlet relation to get the outlet details.
        iii. Join Vehicle and Outlet using OutletID, since each vehicle is assigned to one outlet.
        iv. Return the outlet details together with the registration number, make, and model of each vehicle assigned to that outlet.

D. Define Integrity Constraints

I. Primary Key constraints:
    A. Outlet: outletNo
    B. Vehicle: registrationNo
    C. Client: clientNo
    D. Hire Agreement: hireID
    E. Staff: staffNo

*These attributes are unique and cannot be null

II. Referential Integrity/Foreign Key constraints:
    A. Vehicle.outletNo references Outlet.outletNo
    B. HireAgreement.clientNo references Client.clientNo
    C. HireAgreement.registrationNo references Vehicle.registrationNo
    D. Staff.outletNo references Outlet.outletNo

III. Alternate Key constraints:
    A. Client: Client.DrivingLicenseNumber
        1. The attribute is unique but cannot be null if hiring a vehicle

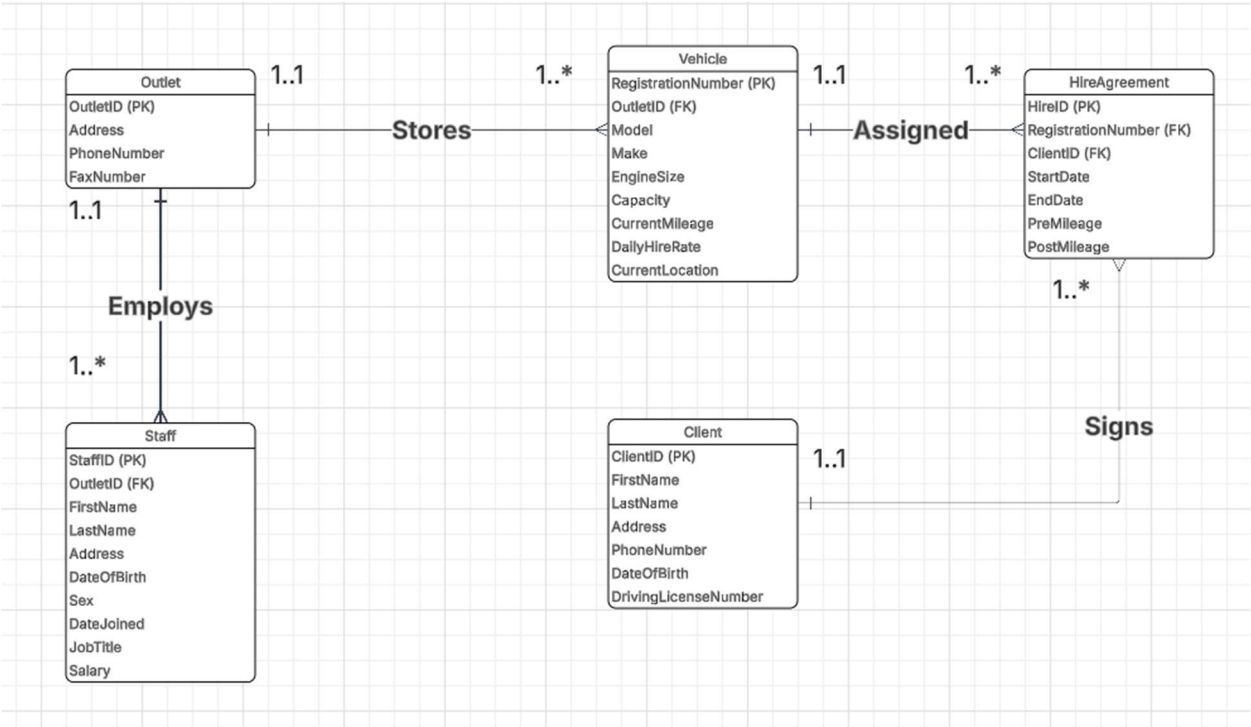
IV. Required Data:
    A. Outlet: OutletID, Address, PhoneNumber
    B. Vehicle: RegistrationNumber, OutletID, Model, Make, EngineSize, Capacity, CurrentMileage, DailyHireRate
    C. Client: ClientID, FirstName, LastName, Address, PhoneNumber, DateOfBirth, DrivingLicenseNumber

D. Staff: StaffID, OutletID, FirstName, LastName, Address, PhoneNumber, DateOfBirth, Sex, DateJoined, JobTitle, Salary
E. HireAgreement: HireID, RegistrationNumber, ClientID, StartDate, EndDate, PreMileage

V. Attribute Domain constraints
   A. EngineSize > 0
   B. Capacity must be a positive integer
   C. CurrentMileage, PreMileage, PostMileage >= 0
   D. DailyHireRate and Salary > 0
   E. Sex must take one of the following values: ‘M’, ‘F’, or ‘Other’
   F. DateOfBirth DateJoined, StartDate, and EndDate must contain valid date values
   G. RegistrationNumber must follow a valid alphanumeric registration format
   H. PhoneNumber must follow a valid phone number format
   I. DrivingLincenseNumber must be unique and follow a valid alphanumeric format

VI. General constraints:
   A. EndDate >= StartDate
   B. PostMileage >= PreMileage
   C. Clients must meet legal driving age based on their date of birth
   D. Each vehicle belongs to one outlet at a time
   E. Each staff member must belong to only one outlet
   F. Each hire agreement must be associated to exactly one client and one vehicle
   G. A vehicle cannot have overlapping hire periods; it can only be associated with the one active hire agreement at a time
   H. Salary and DailyHireRate > 0

E. E-R diagram


Outlet
OutletID (PK)
Address
PhoneNumber
FaxNumber
1..1
Stores
1..*
Vehicle
RegistrationNumber (PK)
OutletID (FK)
Model
Make
EngineSize
Capacity
CurrentMileage
DailyHireRate
CurrentLocation
1..1
Assigned
1..*
HireAgreement
HireID (PK)
RegistrationNumber (FK)
ClientID (FK)
StartDate
EndDate
PreMileage
PostMileage
1..1
Employs
1..*
1..*
Signs
Staff
StaffID (PK)
OutletID (FK)
FirstName
LastName
Address
DateOfBirth
Sex
DateJoined
JobTitle
Salary
Client
ClientID (PK)
FirstName
LastName
Address
PhoneNumber
DateOfBirth
DrivingLicenseNumber
1..1